

## WORKSHEET - NAPLAN Year 9

### Number and Algebra Ratios, Scale and Proportion (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



### Questions

#### 1. Number and Algebra, NAP-61529

Shelly and Carly collect dolls.

The ratio of the number of dolls Shelly owns compared to Carly is 3:2.

Shelley owns 12 dolls.

How many dolls does Carly own?

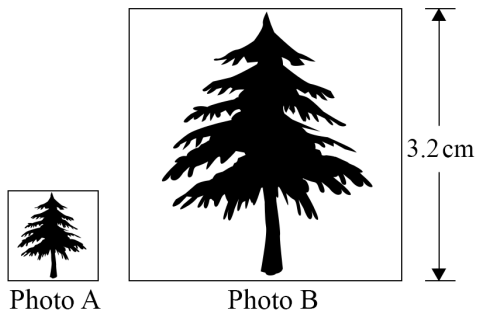
2      6      8      18  
☐   ☐   ☐   ☐

#### 2. Number and Algebra, NAP-55616

Jaunita wanted to print a larger copy of her photo of a tree.

She increased the scale by a factor of 3.

She named the first picture Photo A and the second picture Photo B.



If Photo B is 3.2 cm high, how high is Photo A

0.8 cm      1.12 cm      1.07 cm      9.6 cm  
☐   ☐   ☐   ☐

#### 3. Number and Algebra, NAP-79138

In a hospital, there are 40 nurses and 600 patients.

What is the ratio of nurses to patients?

2:3      1:15      30:2      1:16  
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#### 4. Number and Algebra, NAP-73281

Spiro is making a scale drawing of his house.

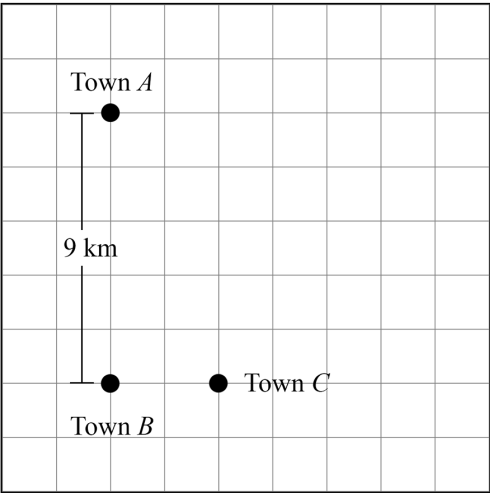
- The height of the garage in the scale model is 6 centimetres.
- The height of Spiro's actual garage is 3 metres.

What does 1 centimetre in Spiro's scale drawing represent in his real house?

1800 cm      200 cm      100 cm      50 cm  
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5.  Measurement and Geometry, NAP-79197

Towns **A**, **B** and **C** are marked on the scale diagram below.

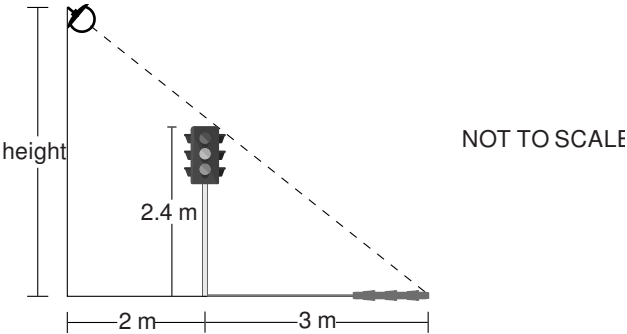


The distance from Town **A** to Town **B** is 9 km.  
What is the distance from Town **B** to Town **C**?

kilometres

6.  Number and Algebra, NAP-61592

A traffic light is 2.4 m tall. Its shadow from a nearby floodlight is 3 m long.



What is the height of the floodlight?

3.6 m

3.8 m

4.0 m

4.2 m

7.  Number and Algebra, NAP-32158

A school has 360 students.  
The ratio of boys to girls is 4 to 6.  
How many boys attend the school?

90

144

216

240

8.  Number and Algebra, NAP-79201

In a class of 28 students, the ratio of boys to girls is 3 : 4.  
Two boys and one girl leave the class. No student joins the class.  
What is the new ratio of boys to girls?

## Worked Solutions

1. Number and Algebra, NAP-61529

Ratio 3:2

Let  $x$  = Number of Carly's dolls

$$\begin{aligned}\frac{x}{12} &= \frac{2}{3} \\ x &= \frac{2 \times 12}{3} \\ &= 8 \text{ dolls}\end{aligned}$$

2. Number and Algebra, NAP-55616

Since Photo B is increased by a factor of 3,

$$\begin{aligned}\frac{\text{Height of Photo B}}{\text{Height of Photo A}} &= 3 \\ \therefore \text{Height of Photo A} &= \frac{\text{Height of Photo B}}{3} \\ &= \frac{3.2}{3} \\ &= 1.07 \text{ cm}\end{aligned}$$

3. Number and Algebra, NAP-79138

$$\begin{aligned}\text{Ratio} &= 40:600 \\ &= 1:15\end{aligned}$$

4. Number and Algebra, NAP-73281

$$\begin{aligned}6 \text{ cm} &= 3 \text{ m} \\ &= 300 \text{ cm} \\ \therefore 1 \text{ cm} &= \frac{300}{6} \\ &= 50 \text{ cm}\end{aligned}$$

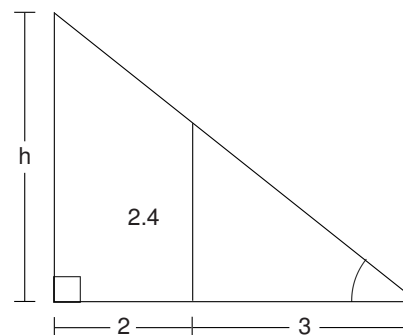
5. Measurement and Geometry, NAP-79197

Let  $d$  = distance of Town B to Town C

Since the diagram is to scale,

$$\begin{aligned}\frac{d}{9} &= \frac{2}{5} \\ \therefore d &= \frac{9 \times 2}{5} \\ &= 3.6 \text{ km}\end{aligned}$$

6. Number and Algebra, NAP-61592



Triangles are similar,

Since similar triangles have sides in similar ratios:

$$\begin{aligned}\frac{h}{5} &= \frac{2.4}{3} \\ \therefore h &= \frac{5 \times 2.4}{3} \\ &= 4 \text{ metres}\end{aligned}$$

7. Number and Algebra, NAP-32158

Ratio is 4 : 6

$$\frac{1}{10} \times 360 = 36$$

$\Rightarrow$  Ratio can also be expressed:

$$4 \times 36 : 6 \times 36$$
$$144 : 216$$

$\therefore$  144 boys attend the school.

8. Number and Algebra, NAP-79201

Original ratio 3 : 4

12 boys : 16 girls

After students leave, new ratio is:

$$10 : 15$$
$$2 : 3$$